

Bootstrap inference in factor-augmented regressions with weak factors

Peiyun Jiang
Waseda University

Takashi Yamagata
University of York

October 2026

Introduction: factor-augmented regressions

- Factor-augmented regressions are widely used in financial and economic research, particularly for predictive regressions in macroeconomics and finance.
- The h -ahead **predictive regression** for y is augmented with a few common factors extracted from a large set of predictors:

$$y_{t+h} = \gamma^{*'} \mathbf{f}_t^* + \beta' \mathbf{w}_t + \epsilon_{t+h}, \quad t = 1, \dots, T,$$

where $\mathbf{f}_t^*$ is an $r \times 1$ vector of *latent* predictive factors and $\mathbf{w}_t$ is a $p \times 1$ vector of observable predictors.

- Since $\mathbf{f}_t^*$ is unobserved, it is replaced by the principal component (PC) estimator $\hat{\mathbf{f}}_t$, extracted from a large $N \times T$ panel:

$$x_{t,i} = \mathbf{b}_i^{*'} \mathbf{f}_t^* + e_{t,i}, \quad t = 1, \dots, T; \quad i = 1, \dots, N.$$

- Stock and Watson (2002); Bai and Ng (2006): use PC factors in predictive regressions; inference on (γ, β) is based on least squares of y_{t+h} on $(\hat{\mathbf{f}}_t', \mathbf{w}_t')'$.

Model: factor-augmented regression

- Matrix form of the predictive regression and factor model:

$$\mathbf{y} = \mathbf{F}^* \gamma^* + \mathbf{W} \beta + \epsilon = \mathbf{Z}^* \delta^* + \epsilon, \quad \mathbf{X} = \mathbf{F}^* \mathbf{B}^{*'} + \mathbf{E},$$

$$\mathbf{Z}^* = (\mathbf{F}^*, \mathbf{W}), \quad \delta^* = (\gamma^{*'}, \beta')'.$$

- Signal eigenvalues, $\lambda_1 > \dots > \lambda_r$, satisfy

$$\lambda_k \asymp N^{\alpha_k}, \quad 1 \geq \alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_r > 0$$

- Model $\mathbf{\Lambda} = \mathbf{D}\mathbf{N}$ with $\mathbf{N} = \text{diag}(N^{\alpha_1}, \dots, N^{\alpha_r})$, $\mathbf{D} = \text{diag}(d_1, \dots, d_r)$.
- Since $\mathbf{F}^*$ is not observable, typically replaced by the PC estimator $\hat{\mathbf{F}}$.
- Consider the conventional approximation (e.g. Bai, 2003)

$$\hat{\mathbf{F}} = \mathbf{F}^* \hat{\mathbf{H}} + o_p(1) \text{ with } \hat{\mathbf{H}} = \mathbf{B}^{*'} \mathbf{B}^* (\mathcal{T}^{-1} \mathbf{F}^{*'} \hat{\mathbf{F}}) \hat{\mathbf{\Lambda}}^{-1}.$$

- Using this approximation, we have $\hat{\mathbf{F}} \hat{\mathbf{H}}^{-1} \gamma^* = \mathbf{F}^* \hat{\mathbf{H}} \hat{\mathbf{H}}^{-1} \gamma^* + o_p(1)$.
- This means regression of $\mathbf{y}$ on $\hat{\mathbf{Z}} = (\hat{\mathbf{F}}, \mathbf{W})$ estimate

$$\delta_{\hat{\mathbf{H}}} := (\gamma^{*'} \hat{\mathbf{H}}^{-1'}, \beta')'$$

Objectives and contributions

Main objective: bootstrap inference for

$$\sqrt{T}(\hat{\delta} - \delta_{\hat{\mathbf{H}}}),$$

and for the alternative rotation targets $\hat{\mathbf{H}}_q$ and $\mathbf{H}$.

Three contributions

1. Rotation-specific bootstrap inference: A direct bootstrap procedure, with first-order validity, for $\mathbf{R} \in \{\hat{\mathbf{H}}, \hat{\mathbf{H}}_q, \mathbf{H}\}$.

2. Bias-corrected inference: Bootstrap studentized analytical and bootstrap bias-corrected estimators; asymptotic pivotality allows scope for bootstrap refinement.

3. Rotation uncertainty: Introduce $\hat{\mathcal{S}}_j$ to assess rotation uncertainty in individual factor coefficients when eigengaps are small.

Model: pseudo-true models and PC estimation

- Jiang et al. (2023): there exists a *unique (up to sign)* signal-dependent rotation $\mathbf{H}(\mathbf{F}^*, \mathbf{B}^*)$ such that $\mathbf{F}^0 := \mathbf{F}^* \mathbf{H}$, $\mathbf{B}^0 := \mathbf{B}^* \mathbf{H}^{-1'}$ satisfy $\mathbf{T}^{-1} \mathbf{F}^{0'} \mathbf{F}^0 = \mathbf{I}_r$ and $\mathbf{B}^{0'} \mathbf{B}^0 = \mathbf{\Lambda}$.
- With $\mathbf{H}$, the model can always be rewritten as **pseudo-true models**

$$\mathbf{y} = \mathbf{F}^0 \gamma^0 + \mathbf{W} \beta + \epsilon = \mathbf{Z}^0 \delta^0 + \epsilon, \quad \mathbf{X} = \mathbf{F}^0 \mathbf{B}^{0'} + \mathbf{E},$$

with $\gamma^0 = \mathbf{H}^{-1} \gamma^*$, $\mathbf{Z}^0 = (\mathbf{F}^0, \mathbf{W})$, $\delta^0 = (\gamma^{0'}, \beta')'$.

- The signal-dependent rotation also satisfies

$$\mathbf{H} = \mathbf{B}^{*'} \mathbf{B}^* (\mathbf{T}^{-1} \mathbf{F}^{*'} \mathbf{F}^0) \mathbf{\Lambda}^{-1} = (\mathbf{T}^{-1} \mathbf{F}^{0'} \mathbf{F}^*)^{-1}.$$

- Regressing $\mathbf{y}$ on $\hat{\mathbf{Z}} = (\hat{\mathbf{F}}, \mathbf{W})$ gives $\hat{\delta} = (\hat{\mathbf{Z}}' \hat{\mathbf{Z}})^{-1} \hat{\mathbf{Z}}' \mathbf{y}$, so $(\hat{\mathbf{F}}, \hat{\delta})$ naturally estimate $(\mathbf{F}^0, \delta^0)$.
- Define the pseudo-true model version of $\hat{\mathbf{H}}$:

$$\tilde{\mathbf{H}} := \mathbf{H}^{-1} \hat{\mathbf{H}} = \mathbf{B}^{0'} \mathbf{B}^0 (\mathbf{T}^{-1} \mathbf{F}^{0'} \hat{\mathbf{F}}) \hat{\mathbf{\Lambda}}^{-1}, \quad \tilde{\mathbf{H}} \xrightarrow{p} \mathbf{I}_r.$$

Then the **key identities** are

$$\mathbf{F}^* \hat{\mathbf{H}} = \mathbf{F}^0 \tilde{\mathbf{H}}, \quad \hat{\mathbf{H}}^{-1} \gamma^* = \tilde{\mathbf{H}}^{-1} \gamma^0.$$

Bootstrap idea: use pseudo-true models

- Consider the pseudo-true model

$$\mathbf{y} = \mathbf{F}^0 \gamma^0 + \mathbf{W} \beta + \epsilon = \mathbf{Z}^0 \delta^0 + \epsilon, \quad \mathbf{X} = \mathbf{F}^0 \mathbf{B}^{0'} + \mathbf{E},$$

with $T^{-1} \mathbf{F}^{0'} \mathbf{F}^0 = \mathbf{I}_r$ and $\mathbf{B}^{0'} \mathbf{B}^0 = \mathbf{\Lambda}$.

- The estimated model by regression $\mathbf{y}$ on $(\hat{\mathbf{F}}, \mathbf{W})$ is

$$\mathbf{y} = \hat{\mathbf{F}} \hat{\gamma} + \mathbf{W} \hat{\beta} + \hat{\epsilon} = \hat{\mathbf{Z}} \hat{\delta} + \hat{\epsilon}, \quad \mathbf{X} = \hat{\mathbf{F}} \hat{\mathbf{B}} + \hat{\mathbf{E}},$$

with $T^{-1} \hat{\mathbf{F}}' \hat{\mathbf{F}} = \mathbf{I}_r$ and $\hat{\mathbf{B}}' \hat{\mathbf{B}} = \hat{\mathbf{\Lambda}}$.

- Thus, regarding the estimates as the bootstrap true parameters $\{\mathbf{F}^{0+}, \mathbf{B}^{0+}, \delta^{0+}\} := \{\hat{\mathbf{F}}, \hat{\mathbf{B}}, \hat{\delta}\}$, with generated bootstrap errors $\{\mathbf{E}^+, \epsilon^+\}$, we can construct a bootstrap sample $\{\mathbf{X}^+, \mathbf{y}^+\}$.
- Also the target of $\hat{\gamma}$ can be directly constructed in the bootstrap due to the identity $\tilde{\mathbf{H}}^{-1} \gamma^0 = \hat{\mathbf{H}}^{-1} \gamma^*$ (in bootstrap $\tilde{\mathbf{H}}^{+1} \gamma^{0+}$).
- Then we can directly construct the primary object, the bootstrap distribution of $\sqrt{T}(\hat{\gamma} - \hat{\mathbf{H}}^{-1} \gamma^*)$

New bootstrap procedure: algorithm

For $b = 1, \dots, B$:

- 1 Generate $\mathbf{X}^{+(b)} = \mathbf{F}^{0+} \mathbf{B}^{0+'} + \mathbf{E}^{+(b)}$ with $\mathbf{F}^{0+} := \hat{\mathbf{F}}$, $\mathbf{B}^{0+} := \hat{\mathbf{B}}$. Obtain PC estimators $(\hat{\mathbf{F}}^{+(b)}, \hat{\mathbf{B}}^{+(b)})$ (signs corrected), and the bootstrap rotation matrix

$$\tilde{\mathbf{H}}^{+(b)} = \mathbf{B}^{0+'} \mathbf{B}^{0+} (T^{-1} \mathbf{F}^{0+'} \hat{\mathbf{F}}^{+(b)}) (\hat{\mathbf{B}}^{+(b)'} \hat{\mathbf{B}}^{+(b)})^{-1}.$$

- 2 Generate $\mathbf{y}^{+(b)} = \mathbf{Z}^{0+} \delta^{0+} + \boldsymbol{\epsilon}^{+(b)}$ with $\delta^{0+} := \hat{\delta}$, $\mathbf{Z}^{0+} = (\mathbf{F}^{0+}, \mathbf{W})$. Compute $\hat{\delta}^{+(b)} = (\hat{\mathbf{Z}}^{+(b)'} \hat{\mathbf{Z}}^{+(b)})^{-1} \hat{\mathbf{Z}}^{+(b)'} \mathbf{y}^{+(b)}$ and the bootstrap target

$$\delta_{\hat{\mathbf{H}}^{+(b)}} = (\gamma^{0+'} \tilde{\mathbf{H}}^{+(b)'}{}^{-1}, \boldsymbol{\beta}^{+'})'$$

to obtain

$$(\hat{\boldsymbol{\Sigma}}_{\delta}^{+(b)})^{-1/2} \sqrt{T} (\hat{\delta}^{+(b)} - \delta_{\hat{\mathbf{H}}^{+(b)}}).$$

- 3 Repeat to build the bootstrap distribution.

Comparison with existing bootstrap

- **Procedure:** Given the same bootstrap errors $(\mathbf{E}^+, \epsilon^+)$, Gonçalves and Perron (2014) use, in our notation,

$$\sqrt{T}(\tilde{\mathbf{H}}^+ \hat{\gamma}^+ - \gamma^{0+}) = \sqrt{T} \tilde{\mathbf{H}}^+ (\hat{\gamma}^+ - \gamma_{\hat{\mathbf{H}}^+}), \quad \tilde{\mathbf{H}}^+ \xrightarrow{p^+} \mathbf{I}_r.$$

The MC evidence below shows that our procedure generally provides a more accurate finite-sample approximation to the bias.

- **Theory:** GP formulate their limiting parameter using the probability-limit rotation $\mathbf{H}_0 = \text{plim } \hat{\mathbf{H}}$ and consider the bootstrap distribution of

$$\sqrt{T}(\hat{\gamma} - \mathbf{H}_0^{-1} \gamma^*),$$

whereas we keep the realized finite-sample rotation-specific target $\gamma_{\hat{\mathbf{H}}}$ explicit.

- This allows us to analyze other rotations, $\hat{\mathbf{H}}_q = (T^{-1} \hat{\mathbf{F}}' \mathbf{F}^*)^{-1}$ and $\mathbf{H}$.

Bootstrap bias-corrected estimator

- Let $\mathcal{R} = \{\hat{\mathbf{H}}, \hat{\mathbf{H}}_q, \mathbf{H}\}$. Jiang et al. (2026) show that, under regularity conditions

$$\sqrt{T}(\hat{\delta} - \delta_{\mathbf{R}}) \xrightarrow{d} N(\mu_{\mathbf{R}}, \Sigma_{\delta}) \text{ for } \mathbf{R} \in \mathcal{R}. \quad (1)$$

- With a consistent bias estimator $\hat{\mu}_{\mathbf{R}} \xrightarrow{p} \mu_{\mathbf{R}}$, we can construct an asymptotically pivotal studentized statistic

$$\hat{\Sigma}_{\delta}^{-1/2} \sqrt{T}(\hat{\delta}^{abc} - \delta_{\mathbf{R}}) \xrightarrow{d} N(\mathbf{0}, \mathbf{I}_{r+p}) \quad (2)$$

where $\sqrt{T}\hat{\delta}^{abc} = \sqrt{T}\hat{\delta} - \hat{\mu}_{\mathbf{R}}$.

- Alternatively, we can use bootstrap bias correction and show:

$$\hat{\Sigma}_{\delta}^{-1/2} \sqrt{T}(\hat{\delta}_{\mathbf{R}}^{bc} - \delta_{\mathbf{R}}) \xrightarrow{d} N(\mathbf{0}, \mathbf{I}_{r+p}), \quad (3)$$

where $\hat{\delta}_{\mathbf{R}}^{bc} = \hat{\delta} - \hat{\mathbf{b}}_{\mathbf{R}}$ with $\hat{\mathbf{b}}_{\mathbf{R}} = B^{-1} \sum_{b=1}^B (\hat{\delta}^{+(b)} - \delta_{\mathbf{R}}^{+(b)})$.

- For testing linear restrictions of $\delta_{\mathbf{R}}$, bootstrapping (2) and (3) potentially yield refinements over bootstrapping (1); Beran (1988)
- bootstrapping (3) requires a second level resampling to recompute its bias correction within each first-level sample.

Second-level bootstrap for the bias-corrected estimator

- 1 For $b = 1, \dots, B$, generate the first-level sample and retain $\hat{\delta}^{+(b)}$, factor estimates, residuals, and $\hat{\Sigma}_{\delta}^{+(b)}$.
- 2 Treat the b th first-level sample as the population of a second-level world. For $\ell = 1, \dots, B_2$, resample with the *same scheme* and re-estimate PC factors and the regression.
- 3 For the fixed target, estimate the conditional bias within the b th first-level world,

$$\hat{\mathbf{b}}_{\mathbf{H}}^{+(b)} = \frac{1}{B_2} \sum_{\ell=1}^{B_2} (\hat{\delta}^{\ddagger(b,\ell)} - \delta^{0\ddagger(b)}), \quad \delta^{0\ddagger(b)} := \hat{\delta}^{+(b)},$$

and form the first-level studentized statistic using $\hat{\delta}^{+(b)} - \hat{\mathbf{b}}_{\mathbf{H}}^{+(b)}$.

- 4 Quantiles over the B first-level statistics deliver bootstrap- t critical values and confidence intervals. The same construction applies to $\hat{\mathbf{H}}$ and $\hat{\mathbf{H}}_q$ with their own rotation targets.

A bootstrap diagnostic for rotation uncertainty

- Consider $\sqrt{T}(\hat{\gamma} - \gamma^0)$ and the decomposition

$$\sqrt{T}(\hat{\gamma} - \gamma^0) = \sqrt{T}(\hat{\gamma} - \gamma_{\hat{\mathbf{H}}_q}) + \sqrt{T}(\gamma_{\hat{\mathbf{H}}_q} - \gamma^0).$$

- The second term tends to a constant vector:

$$\sqrt{T}(\gamma_{\hat{\mathbf{H}}_q} - \gamma^0) = \sqrt{T}(\tilde{\mathbf{H}}_q^{-1} - \mathbf{I}_r)\gamma^0 \xrightarrow{P} 0.5(\mathbf{M}' - \mathbf{M} - \mathbf{1}_{r \times r}) \odot \tilde{\mathbf{G}}\mathbf{H}_0^{-1}\gamma^*,$$

- $\mathbf{M} = (m_{kl})$ is an $r \times r$ matrix with entries

$$m_{kl} = \text{plim} \frac{\lambda_k}{\hat{\lambda}_l - \lambda_k} \text{ for } k \neq l.$$

- Therefore, small eigengaps can amplify uncertainty in individual PC directions, making this term important in finite samples.
- Although this second term is unobservable in the sample, its bootstrap analogue is directly constructible and can be used to assess the variability due to rotation uncertainty.

A bootstrap diagnostic for rotation uncertainty

- In the bootstrap world, $\mathbf{F}^{0+} = \hat{\mathbf{F}}$ is known, so each first-level draw delivers the observable analogue

$$\sqrt{T} \mathbf{g}^{+(b)} := \sqrt{T}(\gamma_{\hat{\mathbf{H}}_q^{+(b)}} - \gamma^{0+}) = \sqrt{T}(\tilde{\mathbf{H}}_q^{+(b)-1} - \mathbf{I}_r) \gamma^{0+}.$$

- For the coefficient on the j th factor, define

$$\hat{S}_j^2 = \text{Var}^+(\sqrt{T} g_j^+) / [\hat{\Sigma}_\delta]_{jj}.$$

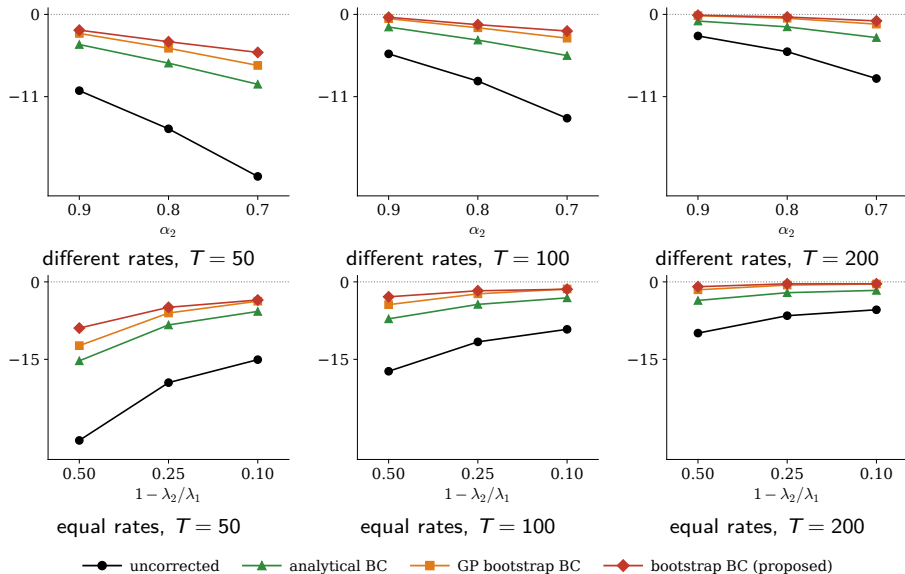
- $\hat{S}_j$ measures variation in the rotation gap relative to sampling variation in the corresponding factor coefficient. Under the maintained eigengap conditions, $\hat{S}_j = o_p(1)$.
- Large finite-sample values indicate that element-wise inference on γ_j^0 is sensitive to uncertainty in the estimated PC direction. $\hat{S}_j$ is a diagnostic, *not* a pretest.

Monte Carlo: design

- $\mathbf{X} = \mathbf{F}^0 \mathbf{B}^{0'} + \mathbf{E}$, $r = 2$; SVD construction gives $T^{-1} \mathbf{F}^{0'} \mathbf{F}^0 = \mathbf{I}_r$, $\mathbf{B}^{0'} \mathbf{B}^0 = \mathbf{D}\mathbf{N}$; $\mathbf{H} = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 2 \end{pmatrix}$.
- Heteroskedastic, skewed idiosyncratic errors; standardized $t(5)$ regression errors; $\rho_{fw} = 0.6$, $p = 2$.
- $T = N \in \{50, 100, 200\}$; 2,000 replications; Mammen wild bootstrap, $B = 299$, $B_2 = 19$; HC0 variances.

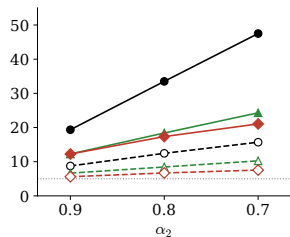
	(α_1, α_2)	(d_1, d_2)	rel. gap $1 - \lambda_2/\lambda_1$	description
D1	(1, 0.9)	(1, 1)	$1 - N^{-0.1}$	distinct rates
D2	(0.9, 0.8)	(1, 1)	$1 - N^{-0.1}$	distinct rates
D3	(0.8, 0.7)	(1, 1)	$1 - N^{-0.1}$	distinct rates
D4	(0.8, 0.8)	(1, 0.50)	0.50	equal rates, wide
D5	(0.8, 0.8)	(1, 0.75)	0.25	equal rates, moderate
D6	(0.8, 0.8)	(1, 0.90)	0.10	equal rates, close

MC results: Bias($\times 100$) vs GP

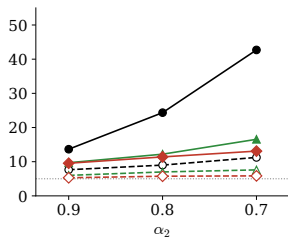


Bias ($\times 100$) of estimators of $\gamma_{2,\hat{H}}$

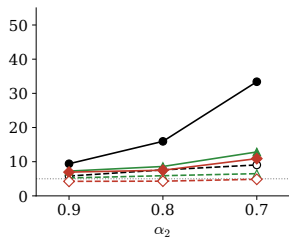
MC results: empirical size (5%), different divergence rates



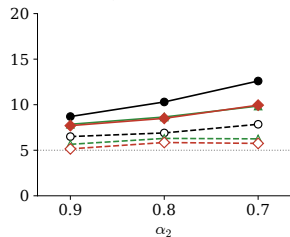
$\gamma_{2, \hat{H}}, T = 50$



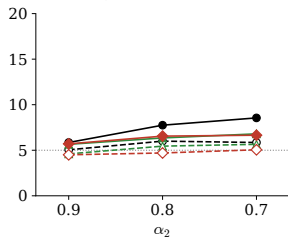
$\gamma_{2, \hat{H}}, T = 100$



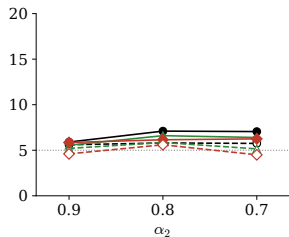
$\gamma_{2, \hat{H}}, T = 200$



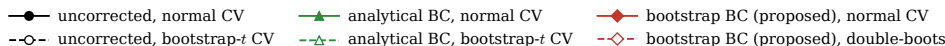
$\beta_1, T = 50$



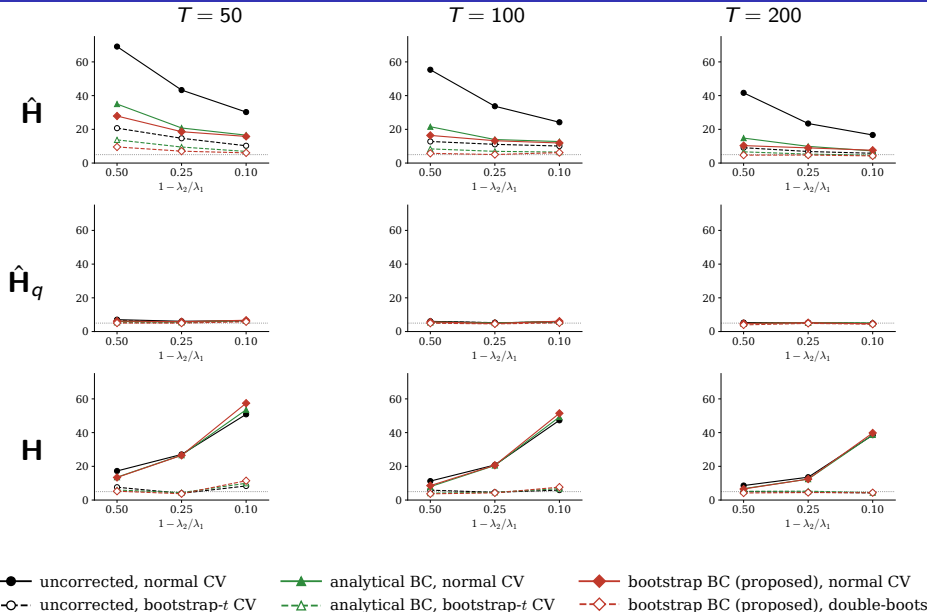
$\beta_1, T = 100$



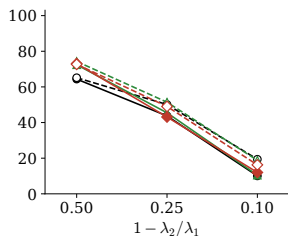
$\beta_1, T = 200$



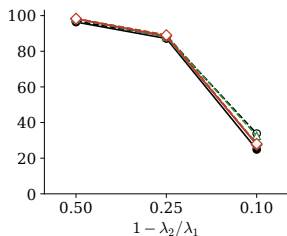
MC: close eigenvalues, empirical size (D4–D6)



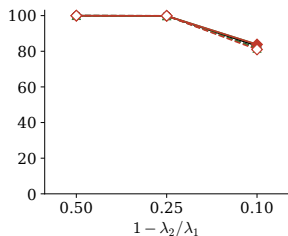
MC: size-adjusted power for the fixed target, D4–D6



$T = 50$



$T = 100$



$T = 200$

—●— uncorrected, normal CV —▲— analytical BC, normal CV —◆— bootstrap BC (proposed), normal CV
- -○- - uncorrected, bootstrap- t CV - -△- - analytical BC, bootstrap- t CV - -◇- - bootstrap BC (proposed), double-bootstrap CV

Size-adjusted power against $\gamma_2^0 \pm 0.4$; relative eigengap 0.50, 0.25, 0.10 runs from wide to close.

MC: rotation-uncertainty diagnostic

	Median $\hat{S}_2$			
	gap \ T	50	100	200
D6 (close)	0.10	2.26	2.47	2.05
D5 (moderate)	0.25	1.31	1.02	0.78
D4 (wide)	0.50	0.61	0.47	0.36

- $\hat{S}_j$ is *coefficient-specific rotation uncertainty relative to conventional sampling uncertainty*; it is not an eigengap statistic itself.
- It rises systematically as the eigengap narrows in these designs: at $T = 100$, median $\hat{S}$ is 2.47 (D6), 1.02 (D5), and 0.47 (D4).
- The diagnostic is not used as a pretest. Its role is to indicate coordinates for which element-wise inference on γ^0 is sensitive to the estimated orientation.

Application: bond return predictability — data and design

- Revisit the application of Jiang et al. (2026), same data and specification $\Rightarrow$ direct comparison of analytical vs. bootstrap inference, holding everything else fixed.
- **Dependent variable:** h -month-ahead log excess return on four-year zero-coupon bonds (Gürkaynak et al., 2011), $h \in \{12, 24\}$.
- **Predictors:** Cochrane and Piazzesi (2005) (CP) factor + $r = 5$ PCs from $N = 65$ FRED-MD macro series (McCracken and Ng, 2016), Groups 1–4 (output/income, labor, housing, consumption–orders–inventories).
- **Subsamples:** 1976M6–2004M1 ($T = 313$) and 2004M2–2024M10 ($T = 249$ for $h = 12$), reflecting the structural change in bond risk premia.
- **Bootstrap dependence:** cross-sectional dependence in factor-model errors is handled by POET whitening/recoloring, while serial dependence is accommodated by block bootstrap resampling.

Application: results, $h = 12$ (target δ^0)

	$\hat{f}_1$	$\hat{f}_2$	$\hat{f}_3$	$\hat{f}_4$	$\hat{f}_5$	CP
<i>1976M6–2004M1 ($T = 313$; rel. gaps 0.64, 0.37, 0.29, 0.04)</i>						
estimate	0.95	0.43	0.08	0.28	1.05	2.41
p : LS, normal	0.22	0.47	0.85	0.25	0.11	0.00
p : analytical BC, normal	0.21	0.48	0.75	0.26	0.05	0.00
p : bootstrap BC, normal	0.22	0.48	0.80	0.28	0.07	0.00
p : bootstrap BC, double boot	0.32	0.59	0.84	0.59	0.23	0.00
$\hat{S}_j$	0.01	0.04	0.18	1.53	0.19	–
<i>2004M2–2024M10 ($T = 249$; rel. gaps 0.66, 0.58, 0.36, 0.11)</i>						
estimate	0.38	1.39	0.03	0.07	0.18	0.14
p : LS, normal	0.10	0.01	0.94	0.80	0.18	0.78
p : analytical BC, normal	0.10	0.01	0.88	0.80	0.14	0.78
p : bootstrap BC, normal	0.10	0.01	0.90	0.79	0.14	0.77
p : bootstrap BC, double boot	0.24	0.07	0.93	0.84	0.21	0.78
$\hat{S}_j$	0.04	0.01	0.09	0.22	0.20	–

Two-sided p -values for $H_0 : \delta_j^0 = 0$; POET-whitened circular-block resampling, $\ell = h$, $B = 999$, $B_2 = 49$. $h = 24$: no coefficient significant under the double bootstrap; same methodological patterns.

Conclusion

- **Direct bootstrap:** valid under possibly heterogeneous weak-factor strengths for $\hat{\mathbf{H}}$, $\hat{\mathbf{H}}_q$, and $\mathbf{H}$; it constructs the bootstrap analogue of the rotation-dependent parameter and uses it directly as the centering parameter, avoiding the additional randomness from pre-rotation.
- **Bias correction and distributional inference:** bootstrap bias-corrected estimators avoid estimating factor-strength rates or analytical bias/eigengap quantities. Their studentized statistics are asymptotically pivotal; the bootstrap bias-corrected estimator requires a second-level bootstrap.
- **Rotation uncertainty:** $\hat{S}$ measures rotation uncertainty relative to conventional sampling uncertainty and is not used as a pretest. Close eigenvalues can make analytical eigengap-based corrections unstable.
- **Application:** the bootstrap procedures and $\hat{S}$ distinguish the robust CP-factor result from macro-factor findings sensitive to rotation uncertainty or to sampling uncertainty under serial dependence.
- **Future work:** analyze rotation uncertainty formally under shrinking eigengaps.

Thank you.